

R Notebook

```
library(tidyverse) # For data manipulation and visualization
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    4.0.0      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.4
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(vegan) # For diversity analyses
```

```
## Loading required package: permute
```

```
library(ggplot2) # For enhanced visualizations
library(dplyr)
library(tidyr)
library(patchwork)
library(kableExtra)
```

```
##
## Attaching package: 'kableExtra'
##
## The following object is masked from 'package:dplyr':
##
##     group_rows
```

```
source("/Users/mattparker/Dropbox/Workspace/moops/manuscript1/helper_utils.r")
knitr::opts_chunk$set(dev = "cairo_pdf")
```

Sample Taxon Reports

```
# List all CSV files in the directory
csv_files <- list.files(path = "/Users/mattparker/Dropbox/Workspace/moops/manuscript1/data/experiment_a",
  seq_metrics = read.csv("/Users/mattparker/Dropbox/Workspace/moops/manuscript1/data/sequencing_summary_m

# Process all CSV files and combine them
all_data <- bind_rows(lapply(csv_files, process_csv_file))

# separate slurry from pond samples
```

```

pond_samples = c("s9","s10","s19","s20")

# Create sample metadata using helper function
sample_metadata = create_sample_metadata(seq_metrics, pond_samples) %>%
  select(sample_name, base_sample, library_prep, experiment, water_control, infection_class, sample_type)

# Merge the taxonomy data with metadata
merged_data <- all_data %>%
  left_join(sample_metadata, by = "sample_name")

# # Create a metadata dataframe to merge with taxon data
# sample_metadata = seq_metrics %>%
#   select(sample_name, water_control, infection_class) %>%
#   mutate(
#     # Library prep: MSSPE if ends with "_M"
#     library_prep = if_else(str_ends(sample_name, "_M"), "MSSPE", "NGS"),
#     #
#     # Remove the _M suffix to get the core sample identifier
#     core = str_remove(sample_name, "_M$"),
#     # Detect experiment 3 first (Cxxx_LPx)
#     is_exp3 = str_detect(core, "C\\d+.*LP\\d+"),
#     #
#     # Base: first sample ID (e.g. S10, PC1)
#     base = str_extract(core, "~[A-Za-z]+\\d+"),
#     #
#     # Append -2 or -3 only if not exp3
#     suffix = case_when(
#       !is_exp3 & str_detect(core, "_2$") ~ "-2",
#       !is_exp3 & str_detect(core, "_3$") ~ "-3",
#       TRUE ~ ""
#     ),
#     #
#     base_sample = tolower(paste0(base, suffix)),
#     #
#     # group slurry vs environmental samples
#     sample_type = if_else(base_sample %in% pond_samples, "environmental", "slurry"),
#     #
#     experiment = case_when(
#       is_exp3 ~ 3,
#       str_detect(core, "_3$") ~ 2,
#       TRUE ~ 1
#     )
#   ) %>%
#   select(sample_name, base_sample, library_prep, experiment, water_control, infection_class, sample_type)
#
# # Merge the taxonomy data with metadata
# merged_data <- all_data %>%
#   left_join(sample_metadata, by = "sample_name")

# Filter to include only species level (tax_level == 1) to avoid duplication
all_data_species <- merged_data %>%
  filter(tax_level == 1) %>%
  filter(water_control == "No") %>%

```

```
filter(infection_class != "Definite") # filter out positive controls
```

Positive Inclusion Criteria

```
species_data_filtered = all_data_species %>%
  filter(nt_rpm >= 1 | nr_rpm >= 1 | nt_count >= 3) %>% # positive inclusion criteria
  filter(nt_alignment_length > 50)
  #filter(sample_type == "slurry") # filtering by sample type
```

Taxonomic Composition

```
# Create a taxa table
taxa_table <- species_data_filtered %>%
  select(tax_id, tax_level, name, genus_tax_id, category, library_prep, nt_rpm, nt_count, base_sample, )
  distinct() %>% # only unique taxa
  filter(!is.na(category)) %>% # filter out species with no aligned category
  filter(category %in% c("archaea", "bacteria", "eukaryota", "viruses")) # because some blank categories are present

phylum_composition_overall <- taxa_table %>%
  group_by(category) %>%
  summarize(total_abundance = sum(nt_rpm, na.rm = TRUE)) %>%
  mutate(relative_abundance = total_abundance / sum(total_abundance) * 100)

# Compare library prep at different taxonomic levels (e.g., phylum level)
phylum_composition <- taxa_table %>%
  group_by(category, library_prep) %>%
  summarize(total_abundance = sum(nt_rpm, na.rm = TRUE)) %>%
  group_by(library_prep) %>%
  mutate(relative_abundance = total_abundance / sum(total_abundance) * 100)
```

`summarise()` has grouped output by 'category'. You can override using the
`.groups` argument.

```
sampletype_composition <- taxa_table %>%
  mutate(
    sample_type = recode(sample_type, `environmental` = "wastewater")
  ) %>%
  group_by(category, sample_type) %>%
  summarize(total_abundance = sum(nt_rpm, na.rm = TRUE)) %>%
  group_by(sample_type) %>%
  mutate(relative_abundance = total_abundance / sum(total_abundance) * 100)
```

`summarise()` has grouped output by 'category'. You can override using the
`.groups` argument.

```
# Visualize composition by library prep method
tax_comp_libprep = ggplot(phylum_composition, aes(x = library_prep, y = relative_abundance, fill = category)) +
  geom_bar(stat = "identity") +
  theme_minimal() +
  labs(title = "a",
       x = "",
```

```

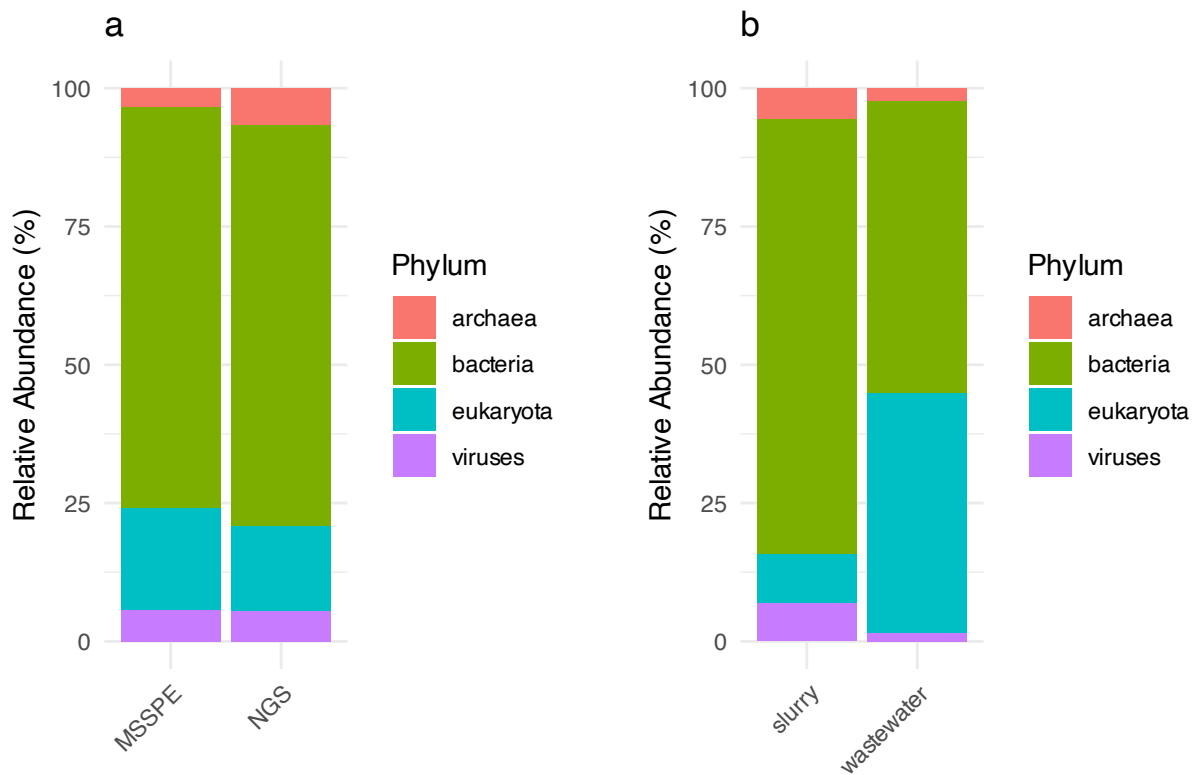
    y = "Relative Abundance (%)",
    fill = "Phylum") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

tax_comp_sampletype = ggplot(sampletype_composition, aes(x = sample_type, y = relative_abundance, fill =
  geom_bar(stat = "identity") +
  theme_minimal() +
  labs(title = "b",
    x = "",
    y = "Relative Abundance (%)",
    fill = "Phylum") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

combined_tax_plots = (tax_comp_libprep | plot_spacer() | tax_comp_sampletype) +
  plot_layout(ncol = 3, widths = c(2, 0.2, 2))

combined_tax_plots

```



Diversity

```
cat("\\captionsetup{labelformat=empty}")
```

```

# Function to compute diversity comparison table
compute_diversity_results <- function(df) {
  species_agg <- df %>%
    group_by(sample_name, tax_id) %>%
    summarise(nt_count = sum(nt_count, na.rm = TRUE), .groups = "drop")

```

```

abundance_matrix <- species_agg %>%
  pivot_wider(names_from = sample_name, values_from = nt_count, values_fill = 0)

abund_mat <- as.matrix(abundance_matrix[,-1])
rownames(abund_mat) <- abundance_matrix$tax_id

diversity_tbl <- data.frame(
  sample_name = colnames(abund_mat),
  Shannon = apply(abund_mat, 2, vegan::diversity, index = "shannon"),
  Simpson = apply(abund_mat, 2, vegan::diversity, index = "simpson"),
  Richness = apply(abund_mat, 2, function(x) sum(x > 0))
) %>%
  left_join(df %>%
    select(sample_name, library_prep, base_sample, sample_type) %>%
    distinct(),
    by = "sample_name")

diversity_tbl <- diversity_tbl %>%
  group_by(base_sample, library_prep) %>%
  summarise(
    Shannon = mean(Shannon),
    Simpson = mean(Simpson),
    Richness = mean(Richness),
    .groups = "drop"
)

diversity_paired = diversity_tbl %>%
  pivot_wider(
    id_cols = base_sample,
    names_from = library_prep,
    values_from = c(Shannon, Simpson, Richness)
  )

metrics <- c("Shannon", "Simpson", "Richness")

results <- lapply(metrics, function(metric) {
  ngs <- diversity_paired[[paste0(metric, "_NGS")]]
  msspe <- diversity_paired[[paste0(metric, "_MSSPE")]]

  test <- wilcox.test(ngs, msspe, paired = TRUE)

  data.frame(
    metric = metric,
    NGS = mean(ngs, na.rm = TRUE),
    MSSPE = mean(msspe, na.rm = TRUE),
    `p-value` = test$p.value
  )
})

bind_rows(results)
}

# ---- Create three datasets ----

```

```

species_slurry <- species_data_filtered %>% filter(sample_type == "slurry")
species_env <- species_data_filtered %>% filter(sample_type == "environmental")

# ---- Compute results ----
tbl_overall <- compute_diversity_results(species_data_filtered) %>% mutate(Type = "Overall")
tbl_slurry <- compute_diversity_results(species_slurry) %>% mutate(Type = "Slurry")
tbl_env <- compute_diversity_results(species_env) %>% mutate(Type = "Wastewater")

# ---- Combine ----
results_all <- bind_rows(tbl_slurry, tbl_env, tbl_overall) %>%
  rename("Diversity Measure" = metric)

# ---- Final kable ----
results_all_kbl = results_all %>%
  kable(
    format = "latex",
    booktabs = TRUE,
    caption = "Comparison of diversity metrics between library preparation methods by sample type",
    digits = 2,
    align = "lrrr", # adjust as needed: text left, numbers right
    format.args = list(big.mark = ",")
  ) %>%
  kable_styling(
    latex_options = c("striped", "hold_position"), # polished journal look
    position = "center",
    full_width = TRUE,
    font_size = 9
  ) %>%
  row_spec(0, bold = TRUE) %>% # strong header
  collapse_rows(columns = 1, latex_hline = "major") # if repeating target names

# 1. Aggregate counts per tax_id per sample
species_agg <- species_data_filtered %>%
  group_by(sample_name, tax_id) %>%
  summarise(nt_count = sum(nt_count, na.rm = TRUE), .groups = "drop")

# 2. Create abundance matrix (taxa x samples)
abundance_matrix <- species_agg %>%
  pivot_wider(
    names_from = sample_name,
    values_from = nt_count,
    values_fill = 0
  )

# Convert to numeric matrix and set rownames
abund_mat <- as.matrix(abundance_matrix[, -1])
rownames(abund_mat) <- abundance_matrix$tax_id

# 3. Compute diversity metrics per sample
diversity_tbl <- data.frame(
  sample_name = colnames(abund_mat),
  Shannon = apply(abund_mat, 2, vegan::diversity, index = "shannon"),
  Simpson = apply(abund_mat, 2, vegan::diversity, index = "simpson"),

```

```

    Richness = apply(abund_mat, 2, function(x) sum(x > 0))
  )

# 4. Add library prep info
diversity_tbl <- diversity_tbl %>%
  left_join(species_data_filtered %>%
    select(sample_name, library_prep, base_sample, sample_type) %>%
    distinct(), by = "sample_name")

# Compare alpha diversity between library prep methods
shannon_libprep_plot = ggplot(diversity_tbl, aes(x = library_prep, y = Shannon, color = library_prep)) +
  geom_boxplot() +
  theme_minimal() +
  labs(title = "a",
        x = "",
        y = "Shannon Diversity Index") +
  theme(legend.position = "none")

simpson_libprep_plot = ggplot(diversity_tbl, aes(x = library_prep, y = Simpson, color = library_prep)) +
  geom_boxplot() +
  theme_minimal() +
  labs(title = "b",
        x = "",
        y = "Simpson Diversity Index") +
  theme(legend.position = "none")

rich_libprep_plot = ggplot(diversity_tbl, aes(x = library_prep, y = Richness, color = library_prep)) +
  geom_boxplot() +
  theme_minimal() +
  labs(title = "c",
        x = "",
        y = "Species Richness") +
  theme(legend.position = "none")

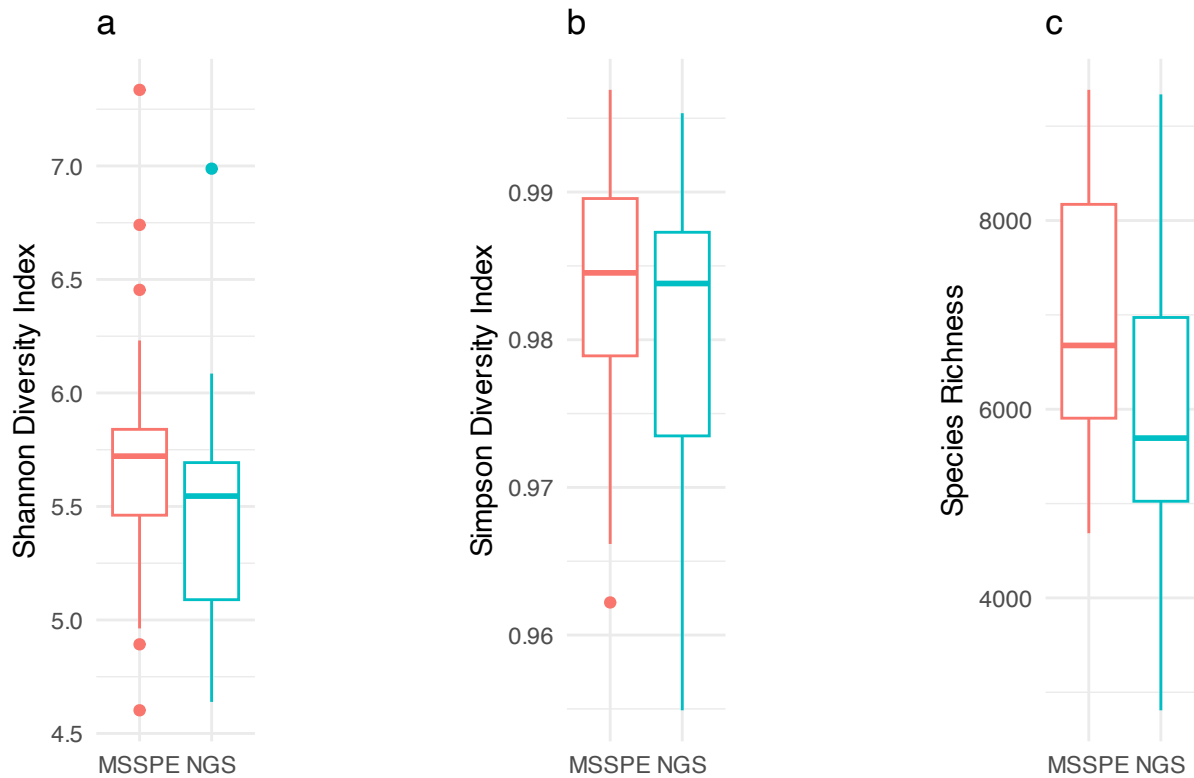
horizontal_space <- plot_spacer()
combined_diversity_plots = (shannon_libprep_plot | horizontal_space | simpson_libprep_plot | horizontal_space | rich_libprep_plot) +
  plot_layout(ncol = 5, widths = c(2, 0.2, 2, 0.2, 2))

## $ncol
## [1] 5
##
## $nrow
## list()
## attr("class")
## [1] "waiver"
##
## $byrow
## list()
## attr("class")
## [1] "waiver"
##
## $widths
## [1] 2.0 0.2 2.0 0.2 2.0

```

```
##
## $heights
## list()
## attr("class")
## [1] "waiver"
##
## $guides
## list()
## attr("class")
## [1] "waiver"
##
## $tag_level
## list()
## attr("class")
## [1] "waiver"
##
## $axes
## list()
## attr("class")
## [1] "waiver"
##
## $axis_titles
## list()
## attr("class")
## [1] "waiver"
##
## $design
## list()
## attr("class")
## [1] "waiver"
##
## attr("class")
## [1] "plot_layout"
```

```
combined_diversity_plots
```

Top Viral Genera

```
# Extract viral category data
virus_data <- species_data_filtered %>%
  filter(category == "viruses") %>%
  filter(is_phage == "false") %>% # filter out phages as marked by CZID
  filter(genus_tax_id != -200)

# Add genus columns to the dataframe
genus_data <- get_genus_info_batch(virus_data$tax_id, batch_size = 100)
```

```
## Processing 687 unique tax IDs out of 3747 total
## Processing batch 1 of 7 - IDs: 12239 to 3060732
## Processing batch 2 of 7 - IDs: 47681 to 3080759
## Processing batch 3 of 7 - IDs: 12242 to 3060074
## Processing batch 4 of 7 - IDs: 10288 to 3118160
## Processing batch 5 of 7 - IDs: 12196 to 3060737
## Processing batch 6 of 7 - IDs: 12205 to 3060826
## Processing batch 7 of 7 - IDs: 10665 to 3052721
## Successfully retrieved genus information for 2701 out of 3747 valid tax IDs
```

```
# Combine with original dataframe
virus_data <- cbind(virus_data, genus_data) %>%
  filter(!genus_name %in% c("Caudoviricetes sp.", "Crassvirales sp.", "Mitoviridae sp.", "uncultured human
```

```
cat("\\captionsetup{labelformat=empty}")
```

```

# get table of only unique species
taxa_table_genus <- virus_data %>%
  select(tax_id, tax_level, name, genus_tax_id, genus_name, category, library_prep, nt_rpm, nt_count, b
  distinct() %>% # only unique taxa
  filter(!is.na(genus_name)) # filter out species with no aligned category

# Calculate viral relative abundance of each viral genus (among all viruses)
genus_composition <- taxa_table_genus %>%
  #group_by(genus_name, library_prep) %>%
  group_by(genus_name) %>%
  summarize(total_abundance = sum(nt_rpm, na.rm = TRUE)) %>%
  #group_by(library_prep) %>%
  mutate(relative_abundance = total_abundance / sum(total_abundance) * 100)

# Calculate mean nt_rpm for each taxon across samples
genus_summary <- virus_data %>%
  group_by(genus_name, library_prep) %>%
  filter(!genus_name %in% c("Alphapapillomavirus", "Nicoomyvirus", "Myrnavirus")) %>% # filter out known
  filter(genus_tax_id != -200) %>%
  summarize(mean_nt_rpm = mean(nt_rpm, na.rm = TRUE),
            total_reads = sum(nt_count, na.rm = TRUE),
            .groups = "drop") %>%
  na.omit() %>%
  arrange(desc(mean_nt_rpm))

# Select top 10 genus names by mean nt_rpm
top_genus_names <- genus_summary %>%
  group_by(genus_name, library_prep) %>%
  summarize(mean_rpm = mean_nt_rpm, .groups = "drop") %>%
  arrange(desc(mean_rpm)) %>%
  slice_head(n = 20) %>%
  pull(genus_name)

# Filter data for top taxa
top_genus_data <- virus_data %>%
  filter(genus_name %in% top_genus_names)
  #na.omit()

# Create overall table with Top Viral Genera
top_genus_summary = top_genus_data %>%
  drop_na(genus_name) %>%
  group_by(genus_name) %>%
  summarise(mean_rpm = mean(nt_rpm), .groups = "drop")
  #slice_head(n=20)

# get frequency of each genus
genus_counts <- virus_data %>%
  drop_na(genus_name) %>%
  count(genus_name, sort = TRUE) %>%
  mutate(genus_name = fct_reorder(genus_name, n))

# get proportion of time genus is detected in each sample
virus_sample_per = virus_data %>%

```

```

drop_na(genus_name) %>%
group_by(genus_name) %>%
summarize(
  perc_of_samples = n_distinct(sample_name) / n_distinct(.$sample_name)
) %>%
arrange(desc(perc_of_samples))

# join back to overall table
top_genus_summary = top_genus_summary %>%
  left_join(genus_counts, by = "genus_name") %>%
  left_join(virus_sample_per, by = "genus_name") %>%
  left_join(genus_composition, by = "genus_name") %>%
  arrange(desc(perc_of_samples))

top_genus_summary_kbl = top_genus_summary %>%
  rename(
    "Genus" = genus_name,
    "rPM" = mean_rpm,
    "Species detections" = n,
    "Percent of Samples" = perc_of_samples,
    "Relative Abundance*" = relative_abundance
  ) %>%
  select(-total_abundance) %>%
  kable(
    format = "latex",
    booktabs = TRUE,
    caption = "Most abundant genera across all samples",
    digits = 2,
    align = "lrrr", # adjust as needed: text left, numbers right
    format.args = list(big.mark = ",")
  ) %>%
  kable_styling(
    latex_options = c("striped", "hold_position"), # polished journal look
    position = "center",
    full_width = TRUE,
    font_size = 9
  ) %>%
  row_spec(0, bold = TRUE) %>% # strong header
  collapse_rows(columns = 1, latex_hline = "major") # if repeating target names

top_genus_summary

```

A tibble: 11 x 6

```

genus_name mean_rpm n perc_of_samples total_abundance relative_abundance
1 Orthopicob~ 161. 338
1 54447. 34.9
2 Rotavirus 397. 144 0.913 57106. 36.6
3 Enterovirus 79.8 79 0.848 6307. 4.04
4 Sapovirus 40.2 47 0.826 1889. 1.21
5 Bocaparvov~ 93.3 113 0.783 10547. 6.76
6 Circovirus 34.7 31 0.674 1077. 0.691
7 Pasivirus 21.8 54 0.652 1175. 0.753
8 Kobuvirus 93.4 16 0.348 1494. 0.958
9 Seadornavi~ 74.1 19 0.348 1408. 0.903
10 Buhlduvirus 22.6 3 0.0652 67.7 0.0434
11 Kahnovirus 28.1

```

```
1 0.0217 28.1 0.0180
```

Virus Overview

```
# get all unique non phage virus genera across all samples  
dim(table(virus_data$genus_name))
```

```
## [1] 189
```

```
# get all unique non phage virus species across all samples  
dim(table(virus_data$name))
```

```
## [1] 684
```

Charting the Viral Community

```
library(forcats)  
library(RColorBrewer)  
library(viridis)
```

```
## Loading required package: viridisLite
```

```
library(scales)
```

```
##
```

```
## Attaching package: 'scales'
```

```
## The following object is masked from 'package:viridis':
```

```
##
```

```
## viridis_pal
```

```
## The following object is masked from 'package:purrr':
```

```
##
```

```
## discard
```

```
## The following object is masked from 'package:readr':
```

```
##
```

```
## col_factor
```

```
library(ggtext)  
library(treemap)  
library(treemapify)
```

```
# Count occurrences of each genus (or use your abundance measure)  
genus_counts <- virus_data %>%  
  drop_na(genus_name) %>%  
  count(genus_name, sort = TRUE) %>%  
  mutate(genus_name = fct_reorder(genus_name, n))
```

```
# For top N genera (adjust N as needed)
```

```

top_genera <- genus_counts %>%
  slice_head(n = 20) # Show top 20 genera

# Percent of samples lollipop
top_genera_lollipop <- ggplot(top_genus_summary, aes(x = perc_of_samples, y = reorder(genus_name, perc_of_samples))) +
  geom_segment(aes(x = 0, xend = perc_of_samples, y = genus_name, yend = genus_name),
    color = "gray70", size = 0.5) +
  geom_point(aes(color = perc_of_samples), size = 3, alpha = 0.8) +
  scale_color_viridis_c(name = "Count", option = "turbo", direction = 1) +
  scale_x_continuous(expand = c(0, 0), labels = comma_format()) +
  labs(
    title = "a",
    x = "Proportion (%)",
    y = "Viral Genus"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(size = 16, face = "bold", hjust = 0.5),
    plot.subtitle = element_text(size = 12, hjust = 0.5, color = "gray60"),
    plot.caption = element_text(size = 10, color = "gray50", hjust = 1),
    axis.title = element_text(size = 12, face = "bold"),
    axis.text.y = element_text(size = 10, face = "italic"),
    axis.text.x = element_text(size = 10),
    legend.position = "none",
    panel.grid.major.y = element_blank(),
    panel.grid.minor = element_blank(),
    plot.margin = margin(20, 20, 20, 20)
  )

```

```

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```

```

# Relative Abundance Treemap
top_genera_treemap <- ggplot(top_genus_summary, aes(area = relative_abundance, fill = relative_abundance))) +
  geom_treemap(alpha = 0.8, color = "white", size = 1) +
  geom_treemap_text(color = "white", place = "centre", size = 10,
    family = "Arial", fontface = "bold") +
  scale_fill_viridis_c(name = "Relative Abundance (%)", option = "turbo", direction = 1) +
  labs(
    title = "b",
    caption = "Relative abundance shown by area size and color"
  ) +
  theme_void(base_size = 12) +
  theme(
    plot.title = element_text(size = 16, face = "bold", hjust = 0.5),
    plot.subtitle = element_text(size = 12, hjust = 0.5, color = "gray60"),
    plot.caption = element_text(size = 10, color = "gray50", hjust = 1),
    legend.position = "right",
    plot.margin = margin(20, 20, 20, 20)
  )

```

```

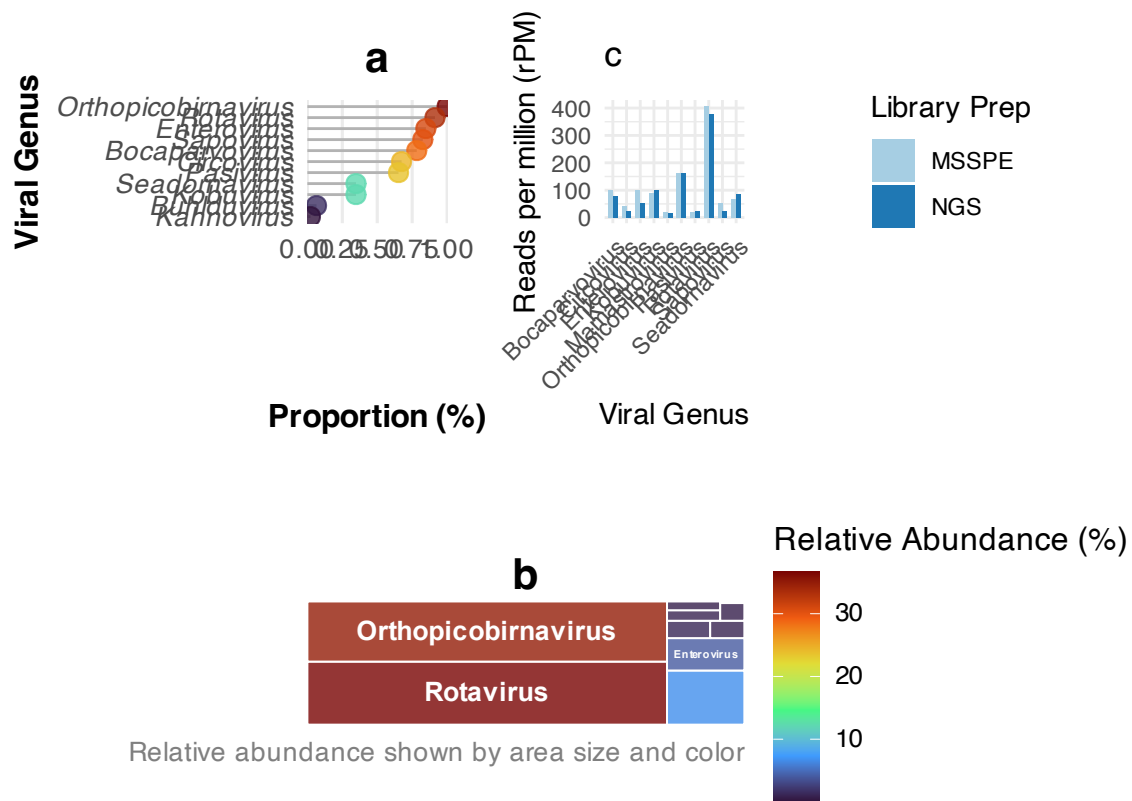
# genus comp by lib prep
top_genus_composition_prep <- taxa_table_genus %>%
  group_by(genus_name, library_prep) %>%
  rename(
    "Library Prep" = library_prep
  ) %>%
  summarize(mean_rpm = mean(nt_rpm, na.rm = TRUE), .groups = 'drop') %>%
  add_count(genus_name) %>%
  filter(n == 2) %>%
  group_by(genus_name) %>%
  mutate(total = mean(mean_rpm)) %>%
  ungroup() %>%
  slice_max(total, n = 20)

#4 Create a boxplot for the top 10 taxa, grouped by control status
genus_boxplot <- ggplot(top_genus_composition_prep, aes(x = genus_name, y = mean_rpm, fill = `Library P
  geom_bar(stat = "identity", position = "dodge", width = 0.7) +
  scale_fill_brewer(palette = "Paired") +
  labs(title = "c",
        x = "Viral Genus",
        y = "Reads per million (rPM)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

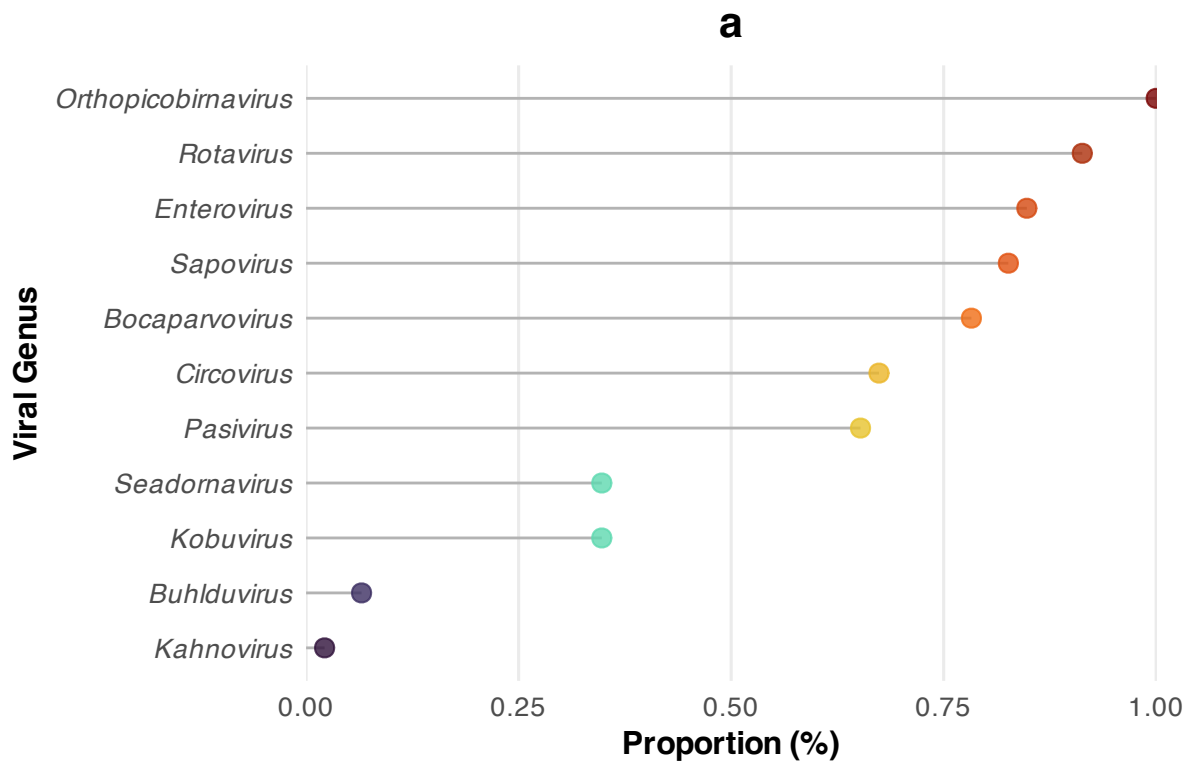
combined_top_genera_plots = (top_genera_lollipop | genus_boxplot) / top_genera_treemap +
  plot_layout(ncol = 1, heights = c(6.5, 6.5))

combined_top_genera_plots

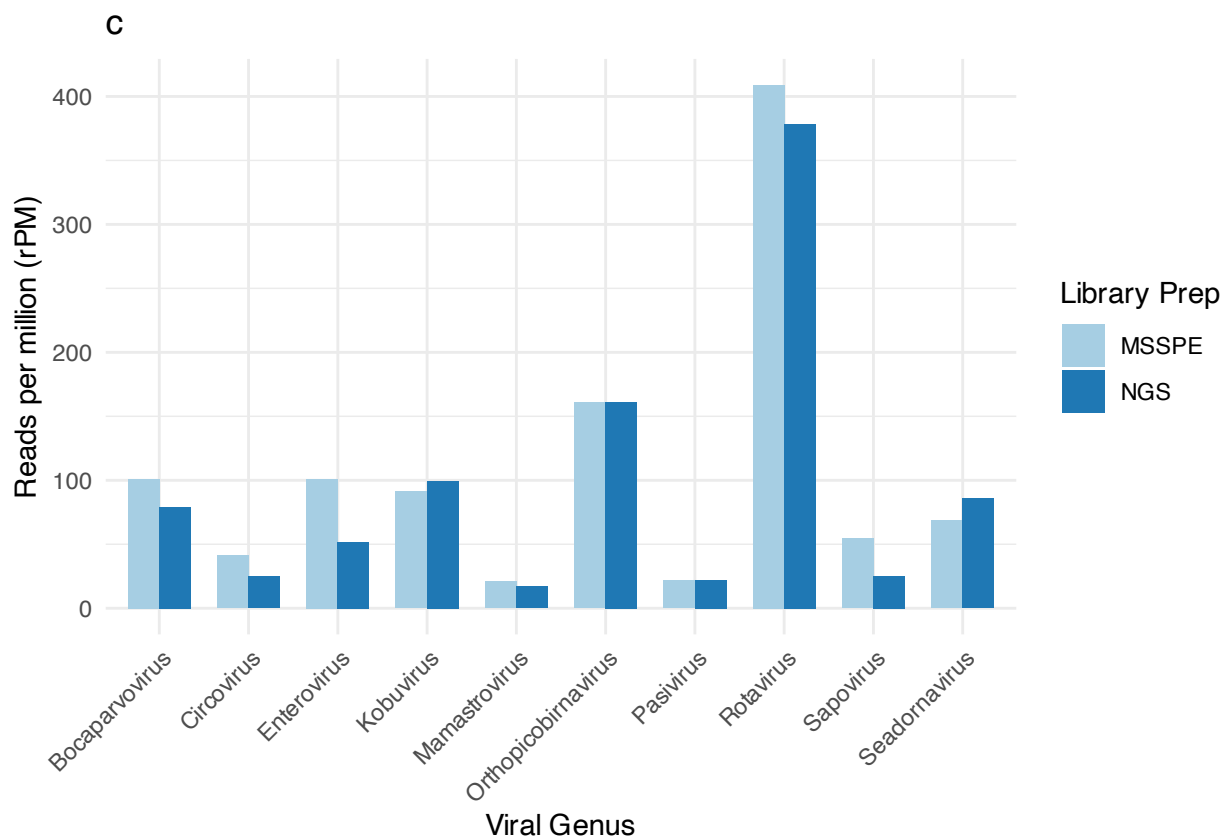
```



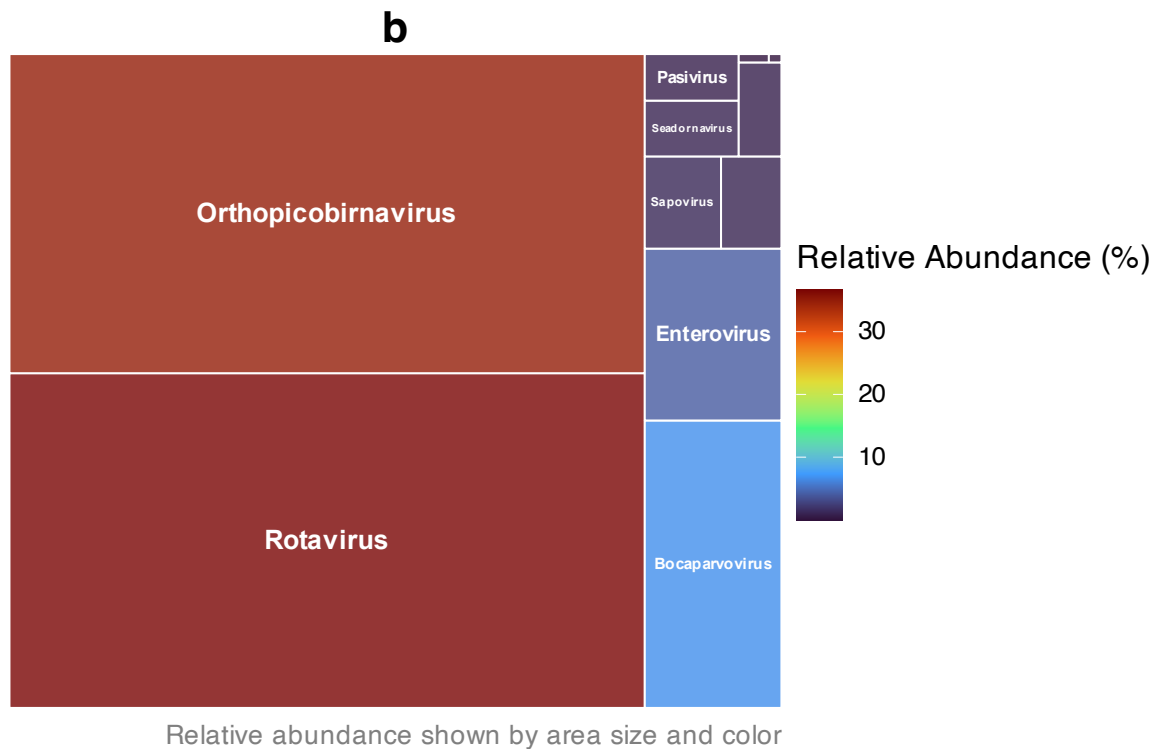
top_genera_lollipop



genus_boxplot



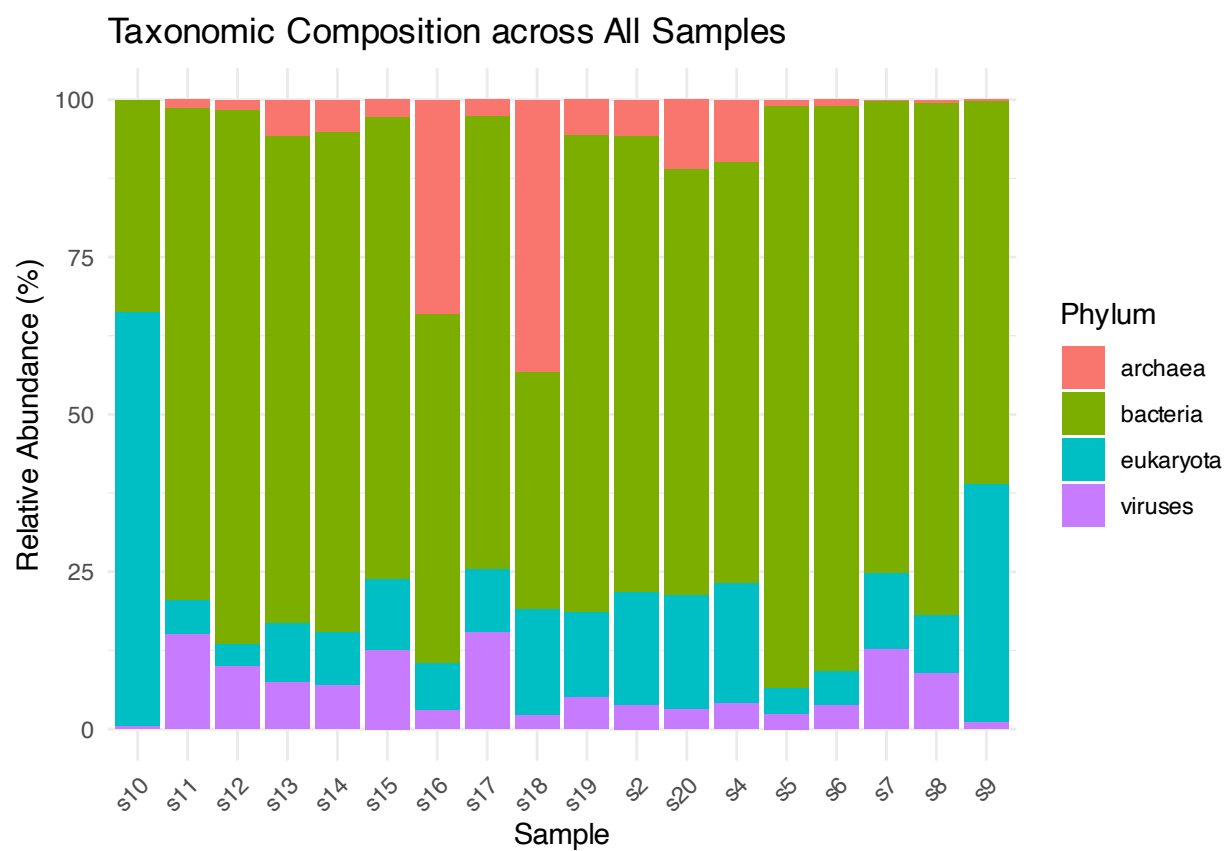
top_genera_treemap



```
# Summarize at different taxonomic levels for each sample
phylum_composition_samples <- taxa_table %>%
  group_by(category, base_sample) %>%
  summarize(total_abundance = sum(nt_rpm, na.rm = TRUE)) %>%
  group_by(base_sample) %>%
  mutate(relative_abundance = total_abundance / sum(total_abundance) * 100)
```

`summarise()` has grouped output by 'category'. You can override using the
`.groups` argument.

```
# Visualize composition by sample
tax_comp_plot_all_samples = ggplot(phylum_composition_samples, aes(x = base_sample, y = relative_abundance)) +
  geom_bar(stat = "identity") +
  theme_minimal() +
  labs(title = "Taxonomic Composition across All Samples",
       x = "Sample",
       y = "Relative Abundance (%)",
       fill = "Phylum") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
tax_comp_plot_all_samples
```



EXPORT ALL FIGURES AND TABLES FTW